

# Business Intelligence – Exercise Sheet: Convolution and K-Means by Hand

Bachelor in Wirtschaftsinformatik, 6th Semester

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## Exercises on Convolution Filters and K-Means Clustering

Exercises 1 to 10 concern manual convolutions. Exercises 11 to 20 concern manual k-means clustering.

For convolution tasks, use the CNN convention: slide the filter as written, without rotating it. Unless stated otherwise, use stride 1 and valid convolution. If same padding is requested, use zero padding. For a grayscale matrix  $X$  and a filter  $K$ , compute

$$Y_{i,j} = \sum_{a=1}^p \sum_{b=1}^q X_{i+a-1,j+b-1} K_{a,b}.$$

For RGB tasks, use the given channel matrices  $R$ ,  $G$ , and  $B$ . For a  $1 \times 1$  RGB filter with weights  $w_R, w_G, w_B$ , compute

$$Y_{i,j} = w_R R_{i,j} + w_G G_{i,j} + w_B B_{i,j}.$$

For RGB filters of size  $2 \times 2$ , sum the contributions of all three channels.

For k-means tasks, use Euclidean distance

$$d(x, m) = \|x - m\|_2$$

and centroid update

$$m_k = \frac{1}{|C_k|} \sum_{x \in C_k} x.$$

You may compare squared Euclidean distances whenever convenient. Break ties in favor of the smaller cluster index. In every k-means exercise, perform

one assignment step, one centroid-update step, and one more assignment step. The first centroid  $m_1^{(0)}$  is intentionally not final, so at least one update is required. When numerical approximations are needed, round to three decimals. Keep fractions whenever convenient. For within-cluster sums of squares, use

$$WCSS = \sum_{k=1}^K \sum_{x \in C_k} \|x - m_k\|_2^2.$$

1. **Ex 1 (20 Points):** A BI system analyzes a small grayscale icon patch with a local contrast filter.

(a) Given

$$X = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 3 & 1 \\ 2 & 1 & 4 \end{bmatrix}, \quad K = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix},$$

compute the top-left output entry

$$Y_{1,1}.$$

(6 Points)

- (b) Compute the full valid convolution output matrix  $Y$ . (8 Points)
- (c) Determine the position of the largest response and explain what this says about the local pattern in the input. (6 Points)

2. **Ex 2 (20 Points):** A dashboard image is processed with a vertical edge detector.

(a) Given

$$X = \begin{bmatrix} 1 & 1 & 1 & 4 \\ 1 & 1 & 1 & 4 \\ 1 & 1 & 1 & 4 \\ 1 & 1 & 1 & 4 \end{bmatrix}, \quad K = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 0 & -1 \\ 1 & 0 & -1 \end{bmatrix},$$

compute

$$Y_{1,1} \quad \text{and} \quad Y_{1,2}.$$

(8 Points)

- (b) Compute the full valid feature map  $Y$ . (6 Points)
- (c) Explain what the sign of the entries means and state where the vertical edge is located. (6 Points)

3. **Ex 3 (20 Points):** A scanner image is processed with a horizontal edge filter.

(a) Given

$$X = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 4 & 4 & 4 & 4 \\ 4 & 4 & 4 & 4 \\ 4 & 4 & 4 & 4 \end{bmatrix}, \quad K = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ -1 & -1 & -1 \end{bmatrix},$$

compute

$$Y_{1,1} \quad \text{and} \quad Y_{2,1}.$$

(8 Points)

(b) Compute the full valid feature map  $Y$ . (6 Points)

(c) Explain why one row of the output becomes zero. (6 Points)

4. **Ex 4 (20 Points):** A BI team applies an averaging filter to blur a simple image patch.

(a) Given

$$X = \begin{bmatrix} 4 & 4 & 0 & 0 \\ 4 & 4 & 0 & 0 \\ 2 & 2 & 8 & 8 \\ 2 & 2 & 8 & 8 \end{bmatrix}, \quad K = \frac{1}{4} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix},$$

compute

$$Y_{1,1} \quad \text{and} \quad Y_{2,2}.$$

(8 Points)

(b) Compute the full valid output matrix  $Y$ . (8 Points)

(c) Explain briefly why the blur filter produces intermediate values near transitions between dark and bright regions. (4 Points)

5. **Ex 5 (20 Points):** A grayscale patch is processed with a Laplacian filter using same padding.

(a) Given

$$X = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \quad K = \begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix},$$

write the zero-padded matrix  $\tilde{X}$ . (4 Points)

(b) Compute

$$Y_{1,1}, \quad Y_{2,2}, \quad Y_{3,3}.$$

(8 Points)

(c) Compute the full same-padded output matrix  $Y$  and identify the position with the strongest positive response. (8 Points)

6. **Ex 6 (20 Points):** A photo patch is sharpened before feature extraction.

(a) Given

$$X = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 3 & 3 & 1 \\ 1 & 3 & 5 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}, \quad K = \begin{bmatrix} 0 & -1 & 0 \\ -1 & 5 & -1 \\ 0 & -1 & 0 \end{bmatrix},$$

compute

$$Y_{1,1} \quad \text{and} \quad Y_{2,2}.$$

(8 Points)

(b) Compute the full valid output matrix  $Y$ . (8 Points)

(c) Identify the strongest sharpened response and explain why it occurs there. (4 Points)

7. **Ex 7 (20 Points):** A document image is processed with an emboss filter.

(a) Given

$$X = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 2 & 4 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad K = \begin{bmatrix} -1 & -1 & 0 \\ -1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix},$$

compute

$$Y_{1,1} \quad \text{and} \quad Y_{2,2}.$$

(8 Points)

(b) Compute the full valid output matrix  $Y$ . (8 Points)

(c) Explain what positive and negative values mean in an emboss-style result. (4 Points)

8. **Ex 8 (20 Points):** A BI pipeline stacks two simple convolution filters.

(a) Given

$$X = \begin{bmatrix} 2 & 2 & 0 \\ 2 & 2 & 0 \\ 1 & 1 & 3 \end{bmatrix}, \quad K_1 = \frac{1}{4} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix},$$

compute the first valid feature map

$$A = X * K_1.$$

(8 Points)

(b) Now apply

$$K_2 = \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix}$$

to  $A$ . Since  $A$  is  $2 \times 2$ , the output is a single scalar. Compute this scalar. (8 Points)

(c) State whether the first blur stage tends to weaken or strengthen local edge contrast before the second filter. (4 Points)

9. **Ex 9 (20 Points):** A simple RGB product image is processed with a warm-color emphasis filter.

(a) The RGB channels are

$$R = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}, \quad G = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 \\ 3 & 0 \end{bmatrix},$$

and the  $1 \times 1$  filter weights are

$$w_R = 1, \quad w_G = 1, \quad w_B = -1.$$

Compute

$$Y_{1,1} \quad \text{and} \quad Y_{2,1}.$$

(8 Points)

(b) Compute the full  $2 \times 2$  output matrix  $Y$ . (8 Points)

(c) Identify which pixels are warm-color dominant in the sense that  $Y_{i,j} > 0$ . (4 Points)

10. **Ex 10 (20 Points):** An RGB patch is processed with one  $2 \times 2 \times 3$  filter.

- (a) The RGB channels are

$$R = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}, \quad G = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix},$$

and the filter channels are

$$K_R = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad K_G = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad K_B = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}.$$

Compute the contribution of the  $R$ -channel to the output scalar.  
(6 Points)

- (b) Compute the contributions of the  $G$ -channel and the  $B$ -channel.  
(8 Points)
- (c) Compute the final output scalar and interpret whether the patch is more red/green-dominated or more blue-dominated with respect to this filter. (6 Points)

11. **Ex 11 (20 Points):** A BI team segments four customers with k-means for  $k = 2$ .

- (a) Given the points

$$A = (1, 1), \quad B = (1, 2), \quad C = (5, 4), \quad D = (6, 5)$$

and initial centroids

$$m_1^{(0)} = (0, 0), \quad m_2^{(0)} = (6, 4),$$

compute the distances from each point to both centroids and assign each point to a cluster. (8 Points)

- (b) Compute the updated centroids

$$m_1^{(1)}, \quad m_2^{(1)}.$$

(6 Points)

- (c) Perform one more assignment step with the updated centroids and decide whether the algorithm has converged. (6 Points)

12. **Ex 12 (20 Points):** A BI analyst clusters one-dimensional order values with k-means for  $k = 2$ .

- (a) Given the data

$$x_1 = 1, \quad x_2 = 2, \quad x_3 = 3, \quad x_4 = 8, \quad x_5 = 9$$

and the initial centroids

$$m_1^{(0)} = 0, \quad m_2^{(0)} = 7,$$

assign all values to the nearest centroid. (6 Points)

- (b) Compute the updated centroids

$$m_1^{(1)}, \quad m_2^{(1)}.$$

(6 Points)

- (c) Perform one more assignment step and determine the final centroids if they change again. (4 Points)
- (d) Compute the total within-cluster sum of squares after convergence. (4 Points)

13. **Ex 13 (20 Points):** A company clusters five products with  $k = 2$  based on two numerical features.

- (a) Given

$$A = (1, 1), \quad B = (2, 1), \quad C = (2, 2), \quad D = (7, 6), \quad E = (8, 7)$$

and initial centroids

$$m_1^{(0)} = (0, 2), \quad m_2^{(0)} = (7, 5),$$

assign all points to the nearest centroid. (6 Points)

- (b) Compute the updated centroids

$$m_1^{(1)}, \quad m_2^{(1)}.$$

(6 Points)

- (c) Perform one more assignment step. (4 Points)
- (d) Compute the total *WCSS* after this second assignment step. (4 Points)

14. **Ex 14 (20 Points):** A BI team forms three market segments with k-means for  $k = 3$ .

(a) Given

$$A = (1, 1), \quad B = (1, 2), \quad C = (5, 1), \quad D = (6, 1), \quad E = (4, 5), \quad F = (5, 6)$$

and initial centroids

$$m_1^{(0)} = (0, 0), \quad m_2^{(0)} = (6, 0), \quad m_3^{(0)} = (3, 6),$$

assign all points to the nearest centroid. (8 Points)

(b) Compute the updated centroids

$$m_1^{(1)}, \quad m_2^{(1)}, \quad m_3^{(1)}.$$

(6 Points)

(c) Perform one more assignment step and state the size of each cluster. (6 Points)

15. **Ex 15 (20 Points):** A simple customer segmentation task uses  $k = 2$ .

(a) Given

$$A = (2, 1), \quad B = (2, 2), \quad C = (3, 2), \quad D = (8, 7), \quad E = (9, 8), \quad F = (9, 9)$$

and initial centroids

$$m_1^{(0)} = (1, 0), \quad m_2^{(0)} = (8, 8),$$

assign all points to the nearest centroid. (6 Points)

(b) Compute the updated centroids

$$m_1^{(1)}, \quad m_2^{(1)}.$$

(6 Points)

(c) Compute the movement lengths

$$\|m_1^{(1)} - m_1^{(0)}\|_2 \quad \text{and} \quad \|m_2^{(1)} - m_2^{(0)}\|_2.$$

(4 Points)

(d) Perform one more assignment step and decide whether the algorithm has converged. (4 Points)

16. **Ex 16 (20 Points):** A BI analyst clusters five users into two groups.



(a) Given

$$A = (1, 3), \quad B = (2, 2), \quad C = (2, 4), \quad D = (7, 7), \quad E = (8, 8)$$

and initial centroids

$$m_1^{(0)} = (0, 4), \quad m_2^{(0)} = (8, 6),$$

assign all points to the nearest centroid. (6 Points)

(b) Compute the updated centroids

$$m_1^{(1)}, \quad m_2^{(1)}.$$

(6 Points)

(c) Perform one more assignment step. (4 Points)

(d) Based on the final centroids, state which cluster corresponds to the lower-valued user group and which one corresponds to the higher-valued user group. (4 Points)

17. **Ex 17 (20 Points):** A set of one-dimensional customer scores is clustered with  $k = 3$ .

(a) Given the data

$$1, \quad 2, \quad 4, \quad 8, \quad 9, \quad 10$$

and initial centroids

$$m_1^{(0)} = 0, \quad m_2^{(0)} = 5, \quad m_3^{(0)} = 11,$$

assign all values to the nearest centroid. (6 Points)

(b) Compute the updated centroids

$$m_1^{(1)}, \quad m_2^{(1)}, \quad m_3^{(1)}.$$

(6 Points)

(c) Perform one more assignment step and indicate whether any point changes cluster. (4 Points)

(d) Compute the new centroids after this second assignment step. (4 Points)

18. **Ex 18 (20 Points):** A BI team clusters six stores into three groups with k-means.

(a) Given

$$A = (1, 1), \quad B = (1, 3), \quad C = (4, 1), \quad D = (5, 2), \quad E = (8, 7), \quad F = (9, 8)$$

and initial centroids

$$m_1^{(0)} = (0, 0), \quad m_2^{(0)} = (5, 1), \quad m_3^{(0)} = (8, 9),$$

assign all points to the nearest centroid. (6 Points)

(b) Compute the updated centroids

$$m_1^{(1)}, \quad m_2^{(1)}, \quad m_3^{(1)}.$$

(6 Points)

(c) Perform one more assignment step. (4 Points)

(d) Compute the total *WCSS* after the second assignment step. (4 Points)

19. **Ex 19 (20 Points):** A BI analyst compares two different initializations for k-means with  $k = 2$  on the same data set.

(a) Use the points

$$A = (1, 1), \quad B = (1, 2), \quad C = (4, 4), \quad D = (6, 5), \quad E = (7, 5).$$

For Initialization A with

$$m_1^{(0)} = (0, 0), \quad m_2^{(0)} = (7, 4),$$

perform one assignment step, compute the updated centroids, and perform one more assignment step. (8 Points)

(b) For Initialization B with

$$n_1^{(0)} = (0, 5), \quad n_2^{(0)} = (7, 0),$$

perform one assignment step, compute the updated centroids, and perform one more assignment step. (8 Points)

(c) Compute the total *WCSS* after the second assignment step for both initializations and decide which initialization is better after one iteration. (4 Points)

20. **Ex 20 (20 Points):** Capstone exercise: a BI system clusters six business objects into three groups with k-means.

(a) Given

$$A = (1, 1), \quad B = (2, 1), \quad C = (5, 4), \quad D = (6, 4), \quad E = (9, 8), \quad F = (10, 9)$$

and initial centroids

$$m_1^{(0)} = (0, 2), \quad m_2^{(0)} = (6, 5), \quad m_3^{(0)} = (10, 7),$$

assign all points to the nearest centroid. (6 Points)

(b) Compute the updated centroids

$$m_1^{(1)}, \quad m_2^{(1)}, \quad m_3^{(1)}.$$

(6 Points)

(c) Perform one more assignment step and decide whether the algorithm has converged. (4 Points)

(d) Compute the final *WCSS* and interpret the three clusters as low-, medium-, and high-valued groups based on their centroid locations. (4 Points)